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Wheelchair docking system and method thereof

Abstract

A wheelchair docking system including a frame having an upper portion and a lower portion, the upper portion being movable relative to the lower portion between a lowered position and a raised position. A coupler mechanism engages a wheelchair during a docking operation. The system also includes a first latching mechanism being movable between a retracted position and a latching position such that the first latching mechanism is spaced from the coupler mechanism. A second latching mechanism moves the upper portion of the frame between its lowered position and raised position. The first latching mechanism is partially retracted by a wheelchair during the docking operation, and the first latching mechanism is biased to its latching position when the coupler mechanism engages the wheelchair.

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Background/Summary

RELATED APPLICATIONS (1) The present disclosure is a continuation-in-part of U.S. patent application Ser. No. 16/909,243, filed Jun. 23, 2020, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/866,064, filed Jun. 25, 2019, entitled “Wheelchair Docking System and Method Thereof,” the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present application relates to a passenger vehicle for transporting one or more passengers, and more particularly to a docking system for releasably coupling a wheelchair to a floor in a vehicle.

BACKGROUND

(2) Automobile manufacturers do not currently mass-produce passenger motor vehicles specifically designed to transport passengers having physical limitations, either as a driver or as a non-driving passenger. Consequently, mass-produced passenger vehicles are modified, or retrofitted, by a number of aftermarket companies dedicated to supplying vehicles to physically limited passengers. Such vehicles can be modified by removing certain parts or structures within a vehicle and replacing those parts with parts specifically designed to accommodate the physically limited passenger. For example, in one configuration, a van or bus is retrofitted with a ramp to enable a physically limited individual using a wheelchair to enter and exit the vehicle without the assistance of another individual.

(3) Other known products for retrofitting a vehicle, such as a van, bus, sport-utility vehicle, or motor coach, include wheel chair lifts, lift platforms, ramps, wheelchair docks, securement systems, and lowered floor surfaces. In some instances, a floor of an original equipment manufacturer (OEM) vehicle is completely removed and replaced, lowered or otherwise modified to accommodate the entry and exit of the physically limited individual through a side door or entrance of the vehicle.

SUMMARY

(4) In a first embodiment of the present disclosure, a wheelchair docking system for being coupled to a floor includes a frame having an upper portion and a lower portion, the upper portion being movable relative to the lower portion between a lowered position and a raised position; a coupler mechanism configured to engage a wheelchair during a docking operation, the coupler mechanism being positioned on the upper portion; a first latching mechanism being movable between a retracted position and a latching position, the first latching mechanism spaced from the coupler mechanism; and a second latching mechanism for moving the upper portion of the frame between its lowered position and raised position; wherein, the first latching mechanism is partially retracted by a wheelchair during the docking operation; wherein, the first latching mechanism is biased to its latching position when the coupler mechanism engages the wheelchair.

- (5) In a first example of this embodiment, the first latching mechanism is biased to its latching position by a spring. In a second example, the first latching mechanism comprises a locking pin. In a third example, an actuator is coupled to the locking pin, the actuator being operably actuated between an extended position and a retracted position to move the locking pin between its latching position and its retracted position.
- (6) In a fourth example, a first scissor assembly is operably coupled between the upper portion and the lower portion; and a second scissor assembly operable coupled between the upper portion and the lower portion, the second scissor assembly being spaced longitudinally from the first scissor assembly. In a fifth example, the first scissor assembly and the second scissor assembly each includes a first leg and a second leg, the first leg and second leg being coupled to one another via a connection pin. In a sixth example, the first leg is disposed outwardly of the second leg.
- (7) In a seventh example, the first leg is coupled to an external location of the upper portion and an internal location of the bottom portion; the second leg is coupled to an internal location of the upper and lower portions. In an eighth example, one end of the first leg is affixed to the lower portion and an opposite end is slidably coupled to the upper portion; one end of the second leg is affixed to the upper portion and an opposite end is slidably coupled to the lower portion. In a ninth example, the second leg of the first scissor assembly is coupled to the second leg of the second scissor assembly via a longitudinal member.
- (8) In a tenth example, the first or second scissor assembly is coupled to a cross member. In an eleventh example, an actuator is coupled to the cross member, the actuator being operably actuated between an extended position and a retracted position to move the cross member longitudinally; wherein, as the cross member moves longitudinally, the upper portion of the frame moves between its lowered position and raised position.
- (9) In another embodiment of the present disclosure, a wheelchair docking system for being coupled to a floor includes a frame having an upper portion and a lower portion, the upper portion being movable relative to the lower portion between a lowered position and a raised position; a coupler mechanism configured to engage a wheelchair during a docking operation, the coupler mechanism being positioned on the upper portion; a first latching mechanism being movable between a retracted position and a latching position, the first latching mechanism spaced from the coupler mechanism; a second latching mechanism for moving the upper portion of the frame between its lowered position and raised position; a first release mechanism for operably controlling movement of the first latching mechanism; and a second release mechanism for operably controlling the second latching mechanism to move the upper portion from its lowered position to its raised position.
- (10) In one example of this embodiment, an actuator is coupled to the first latching mechanism, the actuator being operably actuated between an extended position and a retracted position to move the first latching mechanism between its latching position and its retracted position. In a second example, the first release mechanism includes a user control for communicating with a controller, the controller operably actuating the actuator between its extended and retracted positions; a plate coupled to the first latching mechanism via a pin, the plate being coupled to the actuator; a spring for biasing the first latching mechanism to its latching position; wherein, upon receiving a command from the user control to enable the first release mechanism, the controller operably actuates the actuator which moves the plate for compressing the spring; wherein, as the spring compresses, the first latching mechanism moves from its latching position to its retracted position.
- (11) In another example, the first latching mechanism and the second latching mechanism comprise manually-operable cables. In a further example, the second release mechanism includes a cable operably coupled to a plate having a slot defined therein; a pin disposed within the slot for movement therein from a first position to a second position; an actuator for operably controlling the upper portion between its lowered position and its raised position, the actuator comprising a rod operably coupled to the pin; wherein, in the lowered position, the pin is disposed at a first end of

the slot and the actuator is in a retracted position; wherein, as the cable is pulled, the pin moves from the first end to a second end of the slot, where movement of the pin from the first end to the second end induces the rod to extend in a longitudinal direction; further wherein, movement of the rod in the longitudinal direction induces the upper portion to move from its lowered position to its raised position.

(12) In yet another example, the system includes a first scissor assembly operably coupled between the upper portion and the lower portion; and a second scissor assembly operable coupled between the upper portion and the lower portion, the second scissor assembly being spaced longitudinally from the first scissor assembly; wherein, as the actuator moves from its retracted position to an extended position, the first and second scissor assemblies induce the movement of the upper portion from its lowered position to its raised position.

(13) In a further embodiment of the present disclosure, a wheelchair docking system for being coupled to a floor includes a frame having an upper portion and a lower portion, the upper portion being movable relative to the lower portion between a lowered position and a raised position; a coupler mechanism configured to engage a wheelchair during a docking operation, the coupler mechanism being positioned on the upper portion; a first latching mechanism being movable between a retracted position and a latching position, the first latching mechanism spaced from the coupler mechanism; a second latching mechanism for moving the upper portion of the frame between its lowered position and raised position; a first tether assembly comprising a first tether strap coupled at one end to the lower portion and at an opposite end to the upper portion, the first tether assembly positioned at a rear end of the frame; and a second tether assembly comprising a second tether strap coupled at one end to the lower portion and at an opposite end to the upper portion, the second tether assembly positioned at a front end of the frame.

(14) In an example of this embodiment, the system may include a bracket mounted to the lower portion of the frame; and a pin coupled to the mounting bracket; wherein, the first tether strap is coupled to the pin at the one end.

(15) In an alternative embodiment, a wheelchair docking system for being coupled to a track system coupled to a vehicle floor includes a frame having an upper portion and a lower portion, the upper portion being movable relative to the lower portion between a lowered position and a raised position; a coupler mechanism configured to engage a wheelchair during a docking operation; a bottom panel of the lower portion of the frame comprising a plurality of openings defined therein; a flange integrally formed in the lower portion of the frame, the flange comprising a plurality of openings defined therein; and an adjustable latch coupled to the lower portion via one of the plurality of openings in the flange, the adjustable latch releasably coupled to the track system.

(16) In a first example, the adjustable latch is adjustably coupled to one of the plurality of openings in the flange. In a second example, the plurality of openings is equally spaced and defined laterally in the flange to provide for a laterally adjustable coupling between the lower portion and the track system. In a third example, the plurality of openings in the bottom panel are equally spaced and defined laterally therein. In a fourth example, each of the plurality of openings defined in the flange are aligned longitudinally with at least one of the plurality of openings defined in the bottom panel.

(17) In a fifth example, a body of the adjustable latch with a defined opening; and a fastener coupled to the body via the defined opening, wherein the fastener is disposed through the one of the plurality of openings in the flange and the defined opening in the body for coupling the adjustable latch to the lower portion. In a sixth example, the adjustable latch comprises a tab portion, a first post and a second post, the tab portion being slidable in a generally vertical direction to releasably couple the adjustable latch to the track system. In a seventh example, the tab portion is adjustable between a raised position and a lowered position, where in the lowered position the docking system is coupled to and inhibited from longitudinal movement relative to the track system, and in the raised position the docking system is longitudinally moveable relative to the track system.

(18) In an eighth example, at least one retaining pin includes a neck portion and a retaining end, the

at least one fastener coupling the bottom panel to the track system. In a ninth example, the neck portion is disposed within one of the plurality of openings in the bottom panel.

(19) In another embodiment, a vehicle includes an interior cabin; a vehicle floor located in the cabin; a track system comprising at least one track, the at least one track including a plurality of receptacles and an elongated channel defined therein between a first end and a second end of the at least one track; a wheelchair docking system adjustably coupled to the track system, the wheelchair docking system comprising a frame having an upper portion and a lower portion, a coupler mechanism configured to engage a wheelchair during a docking operation, a bottom panel of the lower portion comprising a plurality of openings defined therein, and a flange integrally formed in the lower portion of the frame; and a latch adjustably coupling the wheelchair docking system to the track system, the latch being coupled to the lower portion via the flange; wherein, each of the plurality of receptacles are equally spaced from an adjacent receptacle; wherein, the flange comprises a plurality of openings defined therein such that the latch is coupled to the docking system via one of the plurality of openings in the flange.

(20) In one example of this embodiment, the track system is mounted directly to a top surface of the vehicle floor. In another example, the plurality of openings in the flange and the plurality of openings in the bottom panel are equally spaced and defined laterally to provide for a laterally adjustable coupling between the docking system and the track system. In yet another example, each of the plurality of openings defined in the flange are longitudinally aligned with at least one of the plurality of openings defined in the bottom panel.

(21) In a further example, the latch comprises a body with an opening defined therein; a tab portion being slidable in a generally vertical direction to releasably couple the latch to the track system; a first post and a second post; and a fastener coupled to the body via the defined opening, wherein the fastener is disposed through the one of the plurality of openings in the flange and the opening in the body for coupling the latch to the docking system. Related thereto, the tab portion is adjustable between a raised position and a lowered position, where in the lowered position the tab portion is disposed within a first receptacle of the plurality of receptacles of the at least one track, and in the raised position the tab portion is released from the first receptacle such that the docking system and latch are longitudinally moveable relative to the track system.

(22) In yet a further example, at least one retaining pin includes a neck portion and a retaining end, the at least one fastener coupling the bottom panel to the track system. Moreover, the neck portion is aligned with and disposed within one of the plurality of openings in the bottom panel and the channel of the at least one track; the retaining end is disposed within one of the receptacles of the plurality of receptacles. Further, related to this embodiment, the at least one track comprises a first track and a second track, the first track and second track being laterally spaced from one another and coupled to the vehicle floor.

(23) In a further embodiment of the present disclosure, a docking assembly for coupling a wheelchair to a floor of a vehicle includes a track system comprising a first track and a second track, the first and second tracks each including a first end, a second end, a plurality of receptacles, and an elongated channel defined therein between the first end and the second end; a wheelchair docking system adjustably coupled to the track system, the wheelchair docking system comprising a frame having an upper portion and a lower portion, a coupler mechanism configured to engage the wheelchair, a bottom panel of the lower portion comprising a plurality of openings defined therein, and a flange integrally formed with the lower portion of the frame; a first fastener adjustably coupling the wheelchair docking system to the first track, the first fastener being coupled to the frame via the flange; and a second fastener adjustably coupling the wheelchair docking system to the second track, the second fastener being coupled to the frame via the flange.

(24) In one example of this embodiment, each of the first and second fasteners includes a tee-bolt, where the tee-bolt comprises an elongated member and at least one stud extending therefrom.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the following description of the embodiments of the disclosure, taken in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 is a front view of a powered wheelchair and docking system in a passenger vehicle;
- (3) FIG. 2 is a partial exploded and perspective view of the wheelchair and docking system of FIG. 1;
- (4) FIG. 3A is a perspective view of a wheelchair docking system and track system for a passenger vehicle in a first configuration;
- (5) FIG. 3B is a perspective view of a wheelchair docking system and track system for a passenger vehicle in a second configuration;
- (6) FIG. 4A is a side view of the wheelchair docking system in the first configuration of FIG. 3A;
- (7) FIG. 4B is a side view of the wheelchair docking system in the second configuration of FIG. 3B;
- (8) FIG. 5A is a side cross-sectional view of the wheelchair docking system in the second configuration;
- (9) FIG. 5B is another side cross-sectional view of the wheelchair docking system in the second configuration;
- (10) FIG. 6 is a partial bottom perspective view of the wheelchair docking system;
- (11) FIG. 7 is a bottom view of the wheelchair docking system;
- (12) FIG. 8 is another bottom view of the wheelchair docking system;
- (13) FIG. 9 is a partial perspective view of a portion of the wheelchair docking system and track system;
- (14) FIG. 9A is a partial perspective view of a release mechanism for adjusting the wheelchair docking system relative to the track system;
- (15) FIG. 10 is a partial exploded perspective view of the portion of the wheelchair docking system and track system of FIG. 9A;
- (16) FIG. 11 is a diagram of a control system for controlling the docking system;
- (17) FIG. 12 is a partial exploded and perspective view of another embodiment of a wheelchair docking system;
- (18) FIG. 13 is a top view of the wheelchair docking system of FIG. 12 with a top portion thereof removed; and
- (19) FIG. 14 is a bottom perspective view of the wheelchair docking system of FIG. 12.

DETAILED DESCRIPTION

- (20) The embodiments of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.
- (21) Referring to FIG. 1 of the present disclosure, a wheelchair **100** is depicted. The wheelchair **100** may include a frame **102** supported by one or more wheels **104**. A brake or anti-tilt/tip mechanism **106** may be located at one or more wheels **104** for slowing down or keeping the wheels **104** from turning, if necessary. The wheelchair **100** may be a powered wheelchair or a manually-operated wheelchair. Any type of wheelchair **100** is applicable to the present disclosure.
- (22) In FIG. 1, the wheelchair **100** is shown located in an interior of a vehicle which has a vehicle floor **108**. The vehicle floor **108** may be the original OEM vehicle floor, or it may be a modified vehicle floor to accommodate a ramp or wheelchair lift assembly. In any event, the wheelchair **100**

may be maneuvered such that the physically limited individual operating or positioned in the wheelchair may be positioned in any location of the vehicle, including at the driver's position of the vehicle.

(23) In conventional vehicle arrangements, a physically limited individual may drive the vehicle so long as the wheelchair is properly latched or connected to the vehicle floor in at least one or two manners. Most conventional wheelchairs are attached to the vehicle via an attendant using four securement belts at each corner of the wheelchair. If the wheelchair user is independent and intends to drive or otherwise operate the vehicle, then most conventional wheelchairs require an aftermarket, conventional wheelchair docking system in which a bolt or other bolt-like feature connected to a bottom of the chair and protruding downward toward the floor. The bolt may then be received by a conventional docking system which is bolted through to the floor. The conventional docking system has a mechanism which receives and latches to the bolt, thereby holding the wheelchair to the vehicle floor. Additional mechanisms may be used to further support and fasten the wheelchair to the vehicle floor.

(24) The conventional wheelchair, however, presents many problems. First, the bolt protrudes downwardly from the wheelchair and leaves very little clearance between the floor and the bolt. Thus, the bolt can often contact objects and the like that the wheelchair would otherwise clear. When the bolt does contact an object, it can cause the wheelchair to tip forward or rearward, or become obstructed with. Alternatively, the object may be dragged by the bolt until it can be cleared from underneath the wheelchair. In either case, it is disadvantageous to have a bolt protruding downwardly from the wheelchair and reducing the clearance between the wheelchair and floor.

(25) In the present disclosure, an improved docking system **112** allows for the wheelchair **100** to have greater clearance between it and the floor **108**. Moreover, the docking system **112** includes a first latching mechanism for coupling to a coupling device **110** on the wheelchair **100**, and a second latching mechanism for coupling the wheelchair **100** to the vehicle floor **108** and preventing it from tilting to the left or right as the vehicle makes a turn. Thus, the present disclosure provides a better connection between the wheelchair **100** and the vehicle floor **108**, and one which is safer over conventional docking systems. Further, the present disclosure provides a track system **114** which allows the docking system **112** to be adjusted longitudinally along the vehicle floor **108** for different sized passengers.

(26) In FIG. 2, for example, the bottom portion of the wheelchair **100** is better shown. Here, the wheelchair **100** has a bottom surface **116** to which the coupling device **110** is connected via one or more fasteners. The coupling device **110** may be a substantially U-shaped bracket **200** formed by a first leg **202** and a second leg **204**. The first and second legs are spaced from one another to define an opening **206** therebetween. In FIG. 2, the opening **206** is oriented towards a front end **208** of the wheelchair **100** rather than a rear end **210**.

(27) The opening **206** in the bracket **200** is configured to engage with the docking system **112**. The docking system **112** may include a frame **216** and a coupler mechanism **218** as shown in FIG. 2. The coupler mechanism **218** may comprise a neck portion **402** (FIG. 4) that extends upwardly from the frame **216** and terminates at a disk-shaped top portion **222**. As the wheelchair **100** is moved into engagement with the docking system **112**, the bracket **200** comes into contact with the coupler mechanism **218**. In particular, the coupler mechanism **218** is received within the opening **206** of the coupling device **110**, and the first leg **202** and second leg **204** are received within a space **400** (FIG. 4A-B) defined between the frame **216** of the docking system **112** and the disk-shaped top portion **222** of the coupler mechanism **218**. In the engaged position, the first leg **202** and second leg **204** may be in close proximity or contact with the neck portion **402** of the coupler mechanism **218**.

(28) To maintain the wheelchair **100** engaged with the docking system **112**, the docking system **112** may further include a retractable locking pin **224**. The locking pin **224** may have an angled surface which comes into contact with a first surface **226** of the bracket **200** causing the locking pin **224** to be pushed downwardly into an opening. Once the bracket **200** clears the locking pin **224**, a spring

512 (FIG. 5) may bias the locking pin **224** to its upward position of FIG. 2. In the upward position, the bracket **200** is retained between the coupler mechanism **218** and the locking pin **224**. This connection between the wheelchair **100** and docking system **112** may establish a first of at least two latching mechanisms of the present disclosure.

(29) The aforementioned track system **114** of the present disclosure is also shown in FIG. 2. Here, the track system **114** may include a first track **212** and a second track **214**. The docking system **112** may be movably coupled to the first and second tracks, which is shown in greater detail in FIGS. 9 and 10. In these illustrated embodiments, a quick-connect track fitting such as an aluminum track fitting may be used for engaging with the tracks. Alternatively, a fastener such as a tee-bolt may be used for engaging the tracks.

(30) The first and second tracks **212**, **214** may be installed at the factory rather than as an after-market part. In doing so, the tracks can be strategically positioned and mounted to the vehicle floor without requiring any drilling through the vehicle floor during installation. As vehicle complexity continues to evolve, the area below the vehicle floor continues to be filled with electrical modules and harnesses, the fuel system, emissions control equipment, and batteries in the case of electric vehicles. As a result, any through-floor drilling runs the risk of damaging any of the systems located underneath the floor. Moreover, any new holes drilled through the vehicle floor can invite moisture and other contaminants into the cabin of the vehicle which can be undesirable for many reasons.

(31) Further, as more electric vehicles become commercially available, battery packs and other electronic equipment may be located underneath the vehicle floor. Any through-floor drilling can damage or even destroy the battery pack or other equipment and create a safety hazard at the same time. Thus, for at least these reasons, it is undesirable to drill through the vehicle floor after the vehicle leaves the factory in order to install the tracks.

(32) In the present disclosure, a method of installing the tracks and docking system in a vehicle includes mounting the tracks to the vehicle floor without requiring any through-floor drilling and/or bolting. The tracks may be mounted in any known way including welding, adhering or fastening. In some instances, the floor may be prepared with holes for assembling the track thereto before any control systems, electrical equipment, fuel lines, exhaust lines, etc. is mounted underneath the floor. In other instances, the tracks are mounting entirely within the cabin and directly to the vehicle floor surface.

(33) 1.sup.st row, front (drivers & passengers), 2.sup.nd row center locations, 3.sup.rd row (in the case of Rear Entry or bus) installation

(34) In FIG. 9, for example, a lower portion **302** of the frame **216** of the docking system **112** is shown. Here, a flange **910** may protrude rearwardly from the lower portion **302** as shown. The docking system **112** spans a gap defined between the first track **212** and second track **214**. In some instances, the gap therebetween may be different or adjustable depending upon the vehicle. Thus, to accommodate different gaps between the first and second tracks, the flange **910** may include a plurality of openings **1004** (FIG. 10) to adjustably couple the docking system **112** to the track system **114**.

(35) The docking system **112** may include a bottom plate or panel **1006** which define a plurality of openings **1008** therein as well. The plurality of openings **1008** are also to accommodate different gaps between the first and second tracks.

(36) Each of the tracks may include a body that has a bottom portion **1020** and a top portion **1022**. The top portion **1022** may have an outer lip that extends outwardly on both sides, as shown in FIG. 10. Moreover, each track defines a plurality of receptacles **1000** configured to receive the docking system **112**. In FIG. 9, for example, the plurality of receptacles **1000** may include a first receptacle **902**, a second receptacle **904**, a third receptacle **906**, a fourth receptacle **908**, and so forth. Each of the plurality of receptacles **1000** is equally spaced from an adjacent receptacle along each longitudinal track. Further, a narrower channel **1024** connects each adjacent receptacle to another

receptacle, as shown in FIG. 10. The channel **1024** may extend from a first end of each track **212**, **214** to an opposite end thereof.

(37) The docking system **112** may be movably coupled to the track system **114** via an adjustable latch **900**. In FIGS. 9 and 9A, the adjustable latch **900** may include a body **918** that defines an opening **1002** for receiving a fastener **916**. In FIG. 10, the fastener **916** may fit through one of the plurality of openings **1004** in the flange **910** and further coupled to the body **918**. For instance, the body **918** may include internal threads to which the fastener **916** may be coupled. The fastener **916** may couple to the body **918** in any conventional manner.

(38) The adjustable latch **900** may include a tab portion **912** which may be slidable in an upward direction **914** as shown in FIG. 9A. As the tab portion **912** is moved in the upward direction **914**, it may be released from being disposed in one of the plurality of receptacles **1000**. As a result, the adjustable latch **900** can be used to move the docking system **112** in a longitudinal direction **1026** relative to the track system **114**. Moreover, as the docking system **112** is moved, a first post **1016** and a second post **1018** on each adjustable latch **900** may slide through the narrow channel **1024** until the tab **912** is repositioned in a different receptacle.

(39) The adjustable latch **900** may be located on the rear of the docking system **112**. At the front of the docking system, a pair of retaining pins may be engaged with the first track **212** and second track **214**. Each retaining pin may include a neck portion **1014** and a retaining end **1012**. A nut **1010** or other fastener may be threadedly coupled to the neck portion **1014** of each retaining pin. Thus, the retaining pin is coupled to the bottom panel **1006** of the docking system. The retaining pin, unlike the adjustable latch **900**, remains coupled to the docking system and may slide in the longitudinal direction **1026** through the channel **1024** as the docking system **112** is adjusted.

Specifically, the neck portion is aligned within the channel **1024** whereas the retaining end is disposed within one of the receptacles. The retaining end is permitted to move within the channel **1024** when the docking system **112** is adjusted longitudinally relative to the first and second tracks.

(40) The openings **1004** in the flange **910** and the openings **1008** in the bottom plate **1006** allow for the docking system **112** to be adjusted side-to-side or perpendicular to the longitudinal direction **1026**. As shown in FIG. 10, the adjustable latches **900** may be positioned in any one of three of the openings **1004** in the flange **910** and the necks **1014** may be positioned in any one of three openings in the bottom plate **1006**. While in the illustrated embodiment there are three openings **1004**, **1008**, in other embodiments there may be two or more openings, i.e., two openings, four openings, five openings, six openings, etc. The side-to-side or lateral adjustability, as well as the fore-and-aft adjustability, can allow a wheelchaired passenger to use the docking system as either a driver or passenger.

(41) Referring now to FIG. 3A, the docking system **112** is shown coupled to the track system **114**. The docking system **112** may include a top portion **300** and a bottom portion **302**. The bottom portion **302** has been described with respect to FIGS. 9-10 above and include the bottom panel **1006** and flange **910**. The top portion **300** forms part of the frame **216** to which the coupler mechanism **218** and locking pin **224** are connected. In FIG. 3A, the docking system **112** is shown in its lowered position **308**, whereas in FIG. 3B the docking system **112** is in its raised position **338**. The docking system **112** is in its raised position **338** when it is not engaged with the wheelchair **100**.

(42) The docking system **112** includes a switch **304** for detecting the presence of the bracket **200** and wheelchair **100**. A wire or other means may electrically couple the switch **304** to a controller **1102** (FIG. 11) for communicating with the driver or physically limited individual that the bracket **200** is coupled to the docking system **112**. For instance, the controller **1102** or a control system **1100** may receive a signal from the switch **304** and display a signal or illuminate a light on a dashboard **1112** of the vehicle indicating the connection.

(43) A second wire or cable **306** is shown in FIG. 3A. This wire or cable **306** may be coupled to a first actuator **334** as shown in FIG. 3B. A control button **1108** located in the vehicle may be

electrically coupled to the actuator **334** via the wire or cable **306**. Alternatively, the controller **1102** or control system **1100** may automatically communicate with the actuator **334** to trigger it between an extended and retracted position. The actuator **334** forms part of the second latching mechanism of the present disclosure.

(44) When the sensor **304** detects that the wheelchair is engaged by the coupler mechanism **218** and locking pin **224**, it may send a signal to a controller **1102** to automatically trigger the actuator **334**. Alternatively, the signal may be displayed on a dashboard **1112** or display screen **1110** in the cab of the vehicle, and the operator may manually trigger the actuator **334**. As the actuator **334** extends and retracts, the top portion **300** may move upwards or downwards relative to the lower or bottom portion **302**. In other words, the actuator **334** may control the movement of the docking system **112** between its raised position **338** of FIG. 3B and its lowered position **308** of FIG. 3A.

(45) The manner in which the docking system **112** moves between its raised and lowered positions will now be described. The docking system **112** may include a front scissor assembly **310** on a front end thereof and a rear scissor assembly **312** on a rear side thereof. Moreover, there may be a front scissor assembly **310** and rear scissor assembly **312** on both the left and right sides of the docking system **112**. The front scissor assembly **310** and rear scissor assembly **312** may include a pair of legs. For example, each assembly may include a first leg **314** and a second leg **316**. The first leg **314** may be disposed outwardly of the second leg **316**. Moreover, the first leg **314** may be coupled to an outside location of the top portion **300** of the docking system **112** and an inside location of the bottom portion **302**. The second leg **316** may be coupled at an inside location of the top and bottom portions of the docking system **112**, as shown in FIG. 3B.

(46) The first leg **314** and second leg **316** may be coupled at an approximate midpoint along the length of each leg. In FIG. 3B, for example, the first leg **314** and second leg **316** may be pivotally coupled via a connection pin **330** or fastener. The first leg **314** and second leg **316** can therefore pivot relative to one another about an axis defined by the connection pin **330**. This is the only coupling point between the first and second legs.

(47) The first leg **314** may be fixedly coupled at one end thereof to the bottom portion **302** of the docking system **112** via a fastener **318**. Thus, the first leg **314** cannot move laterally relative to the bottom portion **302** at this location. At an opposite end, the first leg **314** may be movably coupled to the top portion **300** via a pin **326**. Here, the pin **326** can move within a longitudinal slot **328** defined in the top portion **300** of the docking system **112**. Thus, as the actuator **334** extends and retracts, the first leg **314** remains fixed at one end to the bottom portion **302** via the fastener **318** but moves longitudinally in the slot **328** at an opposite end thereof.

(48) Similarly, the second leg **316** includes two ends. At a first end, the second leg **316** is fixedly coupled to the top portion **300** via a fastener **326**. At an opposite second end, the second leg **316** is movably coupled to the bottom portion **302** via a pin **320**. The pin **320** is able to move longitudinally within a longitudinal slot **322** defined in the bottom portion **302**.

(49) The above-described first and second legs of the front scissor assembly **310** is equally applicable to the front scissor assembly **310** on the opposite side of the docking system **112** as shown in FIG. 3B. Moreover, the rear scissor assembly **312** functions in the same manner. Thus, for sake of brevity, the manner in which the rear scissor assembly **312** operates will not be described.

(50) The rear scissor assembly **312** is also coupled to a cross member **332** as shown in FIG. 3B. In particular, the cross member **332** extends the width of the docking system **112** and is coupled to the pins **320** and second leg **316**. Moreover, the cross member **332** may be coupled to the actuator **334** via a connector **700** (FIG. 7). In FIG. 5A, for example, the connector **700** is shown as a bolt **504** that couples a rod **502** of the actuator **334** to the cross member **332**. As the actuator **334** extends and retracts, the cross member **332** may move longitudinally. Since the cross member **332** is coupled to the second legs **316**, movement of the cross member **332** in a longitudinal direction **500** (FIG. 5) via the actuator **334** may in turn induce the second legs **316** to move longitudinally within the longitudinal slots **322**.

(51) The second leg **316** of the rear scissor assembly **312** may be coupled to the second leg **316** of the front scissor assembly **310** via a longitudinal member **336**. Thus, longitudinal movement of the second leg **316** of the rear scissor assembly **312** is in turn translated into longitudinal movement in the same direction of the second leg **316** of the front scissor assembly **310**. As a result, the docking system **112** is capable of moving between its raised position **338** and lowered position **308** via actuation of the first actuator **334**.

(52) The bottom portion **302** of the docking system **112** may include a first recess **340** and a second recess **342** for receiving the connection pin **330** of the front and rear scissor assemblies in the lowered position **308**.

(53) Once the docking system **112** is in its lowered position **308**, it is better able to maintain the wheelchair **100** from rocking or tilting as the vehicle is making a turn. The locking pin **224** provides a first latching mechanism to connect the wheelchair **100** to the docking system **112**, and the actuation of the docking system **112** to its lowered position functions as a second latching mechanism for holding the wheelchair **100** more securely during vehicle operation.

(54) While the first and second latching mechanisms are able to securely couple the wheelchair **100** to the vehicle floor **108**, there may be an instance where it is desirable to manually release the latching mechanisms. For example, if the vehicle is involved in an accident or there is an emergency, it may be necessary to unlatch the chair from the floor. Alternatively, if the vehicle loses electrical power, it may be necessary to manually release the wheelchair from the docking system **112**. To do so, there are two release systems in place for this.

(55) In FIGS. 5A, 5B, and 6, for example, a first of the release systems is depicted. Here, a first release mechanism **506** is shown for releasing the locking pin **224** and allowing the wheelchair to move away from the docking system **112**. Before describing this release mechanism **506**, however, it is necessary to point out that in FIG. 6 that a second actuator **600** is provided for moving the locking pin **224** between its upward and downward positions. The second actuator **600** may be controlled by a controller **1102** such as the vehicle controller **1114** or any other controller. In one example, a controller **1102** for only controlling the docking system **112** may be provided. In this instance, the controller **1102** may be in communication with the vehicle controller **1114** and/or any other controller of the vehicle (e.g., transmission controller **1118**, engine controller **1116**, etc.) over a communication link such as CAN, J-1939, etc.

(56) The second actuator **600** may be coupled to a plate **604** as shown in FIG. 6. As the actuator **600** is actuated between an extended and retracted position, it induces movement of the plate **604**. As the plate **604** is moved, it is coupled to the locking pin **224** via a pin **602** to move it in a downward position to compress a spring **512**. The actuator **600** may provide sufficient force to the plate **604** to compress the spring **512** and move the locking pin to a retracted position such as shown in FIG. 5A. As the actuator **600** returns to a normal position, the spring **512** may bias the locking pin **224** to its upright position of FIG. 5B. Thus, control of the second actuator **600** allows for releasing the locking pin **224** when desired. Moreover, the locking pin **224** and pin **602** may move in a direction indicated by arrow **610** in FIG. 6.

(57) In some instances, a button or other control **1108** may be in the vehicle to allow the wheelchaired passenger or other individual to control the actuator **600**. The button or control **1108** may be manually triggered, which sends a signal to a controller **1102** which in turn commands the actuator **600** to actuate between its extended and retracted positions. As described above, an alternative embodiment would be for the controller **1102** to automatically detect a condition to release the locking pin **224**. The controller **1102** may include logic, software, or an algorithm to operate from for actuating the first and second actuators of the present disclosure.

(58) The first release mechanism **506** may include a cable or cord **608** of which a user may pull to retract the locking pin **224** from its latched position of FIG. 5B. The cable or cord **608** may be coupled to a cable **508** as shown in FIG. 5A, and one end of the cable **508** may be coupled to the locking pin **224** via a set screw **514** or other fastener. Thus, movement of the cable **508** induces the

locking pin **224** to move downward and compress the spring **512**.

(59) The cable **508** passes through a ferrule **516** as shown in FIG. **5B**. A bracket **510** is further coupled to the cable **508**. The bracket **510** may be similar to or the same as the plate **604**. A ball **606** or other feature may be coupled to the cord **608** and rests against the plate **604** as shown in FIG. **6**. As a user pulls on the cord **608**, it in turn pulls the cable **508** and locking pin **224** downwardly until the locking pin **224** is in the position shown in FIG. **5A**.

(60) A sensor **518** may be provided for detecting a position of the locking pin **224** and communicate this to a controller **1102** or display the position on a dashboard **1112** or other display **1110** in the vehicle. Thus, the operator and/or wheelchair passenger will know the position of the locking pin **224** based on the detection made by the sensor **518**.

(61) The release mechanism **506** is useful to release the locking pin **224** and allow the wheelchair to be disengaged from the docking system **112**. In FIGS. **7** and **8**, a second release mechanism **706** is shown for raising the docking system **112** from its lowered position **308** to its raised position **338**. Here, the second release mechanism **706** may include a safety strap or cable **220** that is coupled to a plate **708** via a fastener **710**.

(62) A spring **344** may be coupled between the plate **708** and the bottom portion **302** of the docking system **112**. In particular, the spring **344** may include a first hook end **702** coupled to the bottom portion **302** and a second hook end **704** coupled to the plate **708**.

(63) The plate **708** may further be coupled to the docking system **112** via a first connector **714**. In addition, the plate **708** may include an L-shaped slot **716** defined therein. A pin **712** may slide or otherwise move within the slot **716**. The pin **712** may be coupled to a rod **806** of the first actuator **334** as shown in FIGS. **7** and **8**. As the pin **712** moves within the slot **716**, the rod **806** may extend or retract.

(64) For example, in FIG. **7**, the docking system **112** may be in its lowered position **308**. The spring **344** is in its free, extended or uncompressed position **718**. The pin **712** is located at a first end of the slot **716** and the actuator rod **806** is in its retracted position. In FIG. **8**, however, the strap or cable **220** may be pulled to achieve a release configuration **800** thereby causing the pin **712** to move within the slot **716** to an opposite end thereof. As it does, the spring **344** is extended along direction **802**. Moreover, as the pin **712** moves to the opposite end of the slot **716**, the pin **712** induces the rod **806** to extend along direction **804** in FIG. **8**. In doing so, the first and second scissor assemblies may raise the docking system **112** an amount equivalent to the length of the slot to relieve down pressure. Thus, in one example, the second release mechanism **706** is capable of transferring the docking system **112** from its lowered position **308** to its raised position **338**. Stated another way, a clamping force on the docking system **112** is in effect relieved. In combination with the first release mechanism **506**, the wheelchair **100** may be manually disengaged and released from the docking system **112** as necessary.

(65) Referring to FIG. **11**, which has been intermittently alluded to in the above description, a control system **1100** for controlling the docking system **112** and the interaction between the docking system **112** and the wheelchair **100** is provided. The control system **1100** may include a controller **1102** which includes a memory unit and processor. The memory is capable of storing logic, algorithms, software, etc. for performing one or more tasks. The memory may further store information, collect data, and receive information from other controllers such as a vehicle controller **1114**, engine controller **1116**, and transmission controller **1118**. The processor or processing unit may be capable of executing the logic, algorithms, software, etc.

(66) In one embodiment, the controller **1102** is a stand-alone controller for controlling the docking system **112**. In another embodiment, the controller **1102** may be the vehicle controller **1114**, the engine controller **1116**, the transmission controller **1118**, or any other controller found on a vehicle. Moreover, the controller **1102** may be remotely located from the vehicle and communicate with the docking system over a wireless communication network such as Wi-Fi.

(67) The controller **1102** may be in communication with a user control **1108** which may be located

in the vehicle. Alternatively, the user control **1108** may be remote from the vehicle. In any event, a user such as the wheelchaired passenger or vehicle operator may send instructions to the controller **1102** by actuating the user control **1108**.

(68) In turn, the controller **1102** may communicate with the user by displaying a signal, data, information, instructions, etc. via a display **1110** or dashboard **1112**. The display **1110** or dashboard **1112** may be located in the vehicle. The display **1110** may be a computer display. The signal may be communicated by illuminating a light in the vehicle to alert the user that the docking station **112** is engaged with the wheelchair **110** or vice versa. Other types of signals are also possible.

(69) The controller **1102** may be in communication with the first actuator **334** and second actuator **600** of the docking system **112**. In this manner, the controller **1102** may command either or both actuators to extend or retract. This may be based on a user command via the user control **1108**, or it may be part of the control logic, algorithms, software, etc. executed by the processor of the controller **1102**.

(70) The controller **1102** may receive signals from one or more sensors. For example, the sensor **518** may detect the position of the locking pin **224** and communicate this position to the controller **1102**. A second sensor **1104** may detect a position of the first actuator **334** and/or second actuator **600**. For example, the controller **1102** may command the actuator **334** to extend by a desired amount. The second sensor **1104** may detect how much the actuator **334** has extended and communicate the same to the controller **1102**. In this way, the controller **1102** receives feedback from the sensor **1104** and can further adjust its commands to either actuator.

(71) The control system **1100** may include a third sensor **1106** which may be positioned on the docking system **112** and is able to detect an oncoming wheelchair **100**. The third sensor **1106** may be a proximity sensor, Hall Effect sensor, or any other type of sensor. The third sensor **1106** may detect a height or clearance between a bracket **200** on the approaching wheelchair **100** and communicate the same to the controller **1102**. In turn, the controller **1102** may actuate the first actuator to cause the docking system **112** to move upwards or downwards based on the detected clearance by the third sensor **1106**. In doing so, the second sensor **1104** can detect how far and in what direction the actuator **334** moves in order to determine if the actuator **334** responded correctly based on the instruction from the controller **1102**. A fourth sensor (not shown) may detect the height of the coupler mechanism **218** relative to the vehicle floor **108** and communicate the same to the controller **1102**. Thus, the controller **1102** is able to receive signals indicative of an approaching wheelchair **100**, the desired height of the docking system **112** for receiving the wheelchair **100**, the actual height of the docking system **112**, and the responsiveness of the first actuator **334** for adjusting the height of the docking system **112**.

(72) Additional control logic or algorithms may be performed by the control system **1100** for docking the wheelchair **100** to the docking system **112**. One or more controllers may execute the control logic or algorithms. In a further embodiment, the wheelchair may include a controller or transmitter for communicating with the control system **1100**. In this manner, the transmitter or controller on the wheelchair may alert the controller **1102** or third sensor **1106** of its approach.

(73) Referring now to FIGS. **12** and **13**, a different embodiment of a wheelchair docking system **1200** is illustrated. For sake of brevity, features in the embodiment of FIGS. **12** and **13** that remain unchanged from the embodiments in FIGS. **1-11** include the same reference number. For this reason, only the features that have changed between embodiments will be addressed.

(74) In FIG. **12**, the wheelchair docking system **1200** may include a lower protective shroud **1202** and an upper protective shroud **1204**. The lower protective shroud **1202** may be formed of a plastic material formed by any known process such as injection molding. The upper protective shroud **1204** may be formed of sheet metal or similar material. The pair of shrouds provide additional safety and aesthetic benefits to the wheelchair docking system **1200**.

(75) The docking system **1200** may also include a plurality of tether assemblies for increased structural integrity and improvement. For instance, a rear tether assembly **1206** is depicted in FIGS.

12 and **13** having a rear load tether strap **1208**. The rear load tether strap **1208** may be coupled at one end to a pin **1302** which is affixed to a mounting bracket **1300**. The mounting bracket **1300** may be welded or otherwise coupled to the lower portion **302** of the system **1200**. In at least one embodiment, the pin **1302** may pivot or rotate about an axis within the bracket **1300**.

(76) The rear load tether strap **1208** may be coupled at its opposite end to the top portion **300** of the system. The rear tether assembly **1206** may be approximately centrally located as shown in FIGS.

12 and **13**. Moreover, the top portion **300** may be a cast material for improved structural integrity.

(77) The wheelchair docking system **1200** may also include a pair of front tether assemblies. A first front tether assembly **1210** may be located at a first front corner and a second front tether assembly **1212** may be located at a second front corner. Each tether assembly may include a tether strap similar to that of the rear tether assembly **1206**. For instance, the first front tether assembly **1210** may include a tether strap **1304** and the second front tether assembly **1212** may include a tether strap **1306**. Each tether strap **1304**, **1306** may be coupled at one end to the lower portion **302** and at the opposite end to the top portion **302**.

(78) The docking system **1200** may also include an upper and lower gussets. The lower gusset **1308** is located at the front scissor assembly **310**. The upper gusset, which is not shown in FIGS. **12** and **13**, is also located at the front scissor assembly **310**. Each gusset may be welded to the system. In particular, the lower gusset **1308** may be welded to the lower portion **302**, whereas the upper gusset may be welded to the top portion **300**.

(79) As also shown in FIG. **13**, a plurality of fasteners or stud bolts may be used to further couple the wheelchair docking system **1200** to the tracks **212**, **214**. As shown, a pair of stud or tee-bolts **1310** is shown at approximately each front corner of the system **1200** to add further structural integrity to the overall system.

(80) Moreover, rather than the adjustable latches of FIG. **9**, a different or alternative design may be implemented for movably coupling the docking system **112** to the track system **114**. In FIGS. **13** and **14**, for example, a tee-bolt **1312** may be used. The tee-bolt **1312** may include an elongated member **1400** configured to slide or move within the channel **1024** of each track. The elongated member **1400** may include a narrow raised portion **1318** that is also capable of moving within the narrow channel **1024**. Further, the tee-bolt **1312** may comprise one or more studs **1320** extending from the elongated portion **1400**. In FIG. **13**, each tee-bolt **1321** includes two studs **1318** extending in a substantially perpendicular direction from the elongated member **1400**.

(81) A plurality of fasteners may couple to the studs for securing the tee-bolt to the docking station **112**. For example, a first bolt **1314** and a second bolt **1316** are shown in FIG. **13** for each tee-bolt **1312**. These bolts or fasteners rigidly couple each stud of the respective tee-bolt **1312** to the docking system **112** to allow the docking system **112** to move smoothly within each track.

(82) As shown, the elongated members **1400** may include a substantially flat bottom surface to allow it to smoothly slide or move within the tracks. The bottom surface of each of the elongated members **1400** may be approximately parallel to a bottom surface of the lower portion **302** of the frame **216**, as shown in FIG. **14**. This can help ensure smooth movement of the docking system **112** along the tracks.

(83) While exemplary embodiments incorporating the principles of the present disclosure have been disclosed herein, the present disclosure is not limited to the disclosed embodiments. Instead, this application is intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A wheelchair docking system for being coupled to a track system coupled to a vehicle floor, comprising: a frame; a coupler mechanism configured to engage a wheelchair as the wheelchair approaches the wheelchair docking system during a docking operation; an adjustable latch coupled to the frame, the adjustable latch configured to be releasably coupled to the track system; wherein the frame comprises a flange with a plurality of openings and wherein the adjustable latch is adjustably coupled to one of the plurality of openings in the flange.
2. The docking system of claim 1, wherein the plurality of openings is equally spaced and defined laterally in the flange to provide for a laterally adjustable coupling between the frame and the track system.
3. The docking system of claim 2, wherein a bottom panel of the frame comprises a plurality of openings defined therein and wherein the plurality of openings in the bottom panel is equally spaced and defined laterally therein.
4. The docking system of claim 3, wherein each of the plurality of openings defined in the flange is aligned longitudinally with at least one of the plurality of openings defined in the bottom panel.
5. The docking system of claim 4, further comprising: a body of the adjustable latch with a defined opening; and a fastener coupled to the body via the defined opening, wherein the fastener is disposed through the one of the plurality of openings in the flange and the defined opening in the body for coupling the adjustable latch to the frame.
6. The docking system of claim 1, wherein the adjustable latch comprises a tab portion, a first post and a second post, the tab portion being slidable in a generally vertical direction to releasably couple the adjustable latch to the track system.
7. The docking system of claim 6, wherein the tab portion is adjustable between a raised position and a lowered position, where in the lowered position the docking system is coupled to and inhibited from longitudinal movement relative to the track system, and in the raised position the docking system is longitudinally moveable relative to the track system.
8. The docking system of claim 1, further comprising at least one retaining pin comprising a neck portion and a retaining end, the at least one retaining pin coupling the frame to the track system.
9. The docking system of claim 8, wherein the neck portion is disposed within one of a plurality of openings in a bottom panel.
10. The docking system of claim 1 in combination with a vehicle, wherein: the vehicle comprises an interior cabin; the vehicle floor is located in the cabin; the track system comprises at least one track, the at least one track including a plurality of receptacles and an elongated channel defined therein between a first end and a second end of the at least one track; the wheelchair docking system is adjustably coupled to the track system.
11. The combination of claim 10, wherein the track system is mounted directly to a top surface of the vehicle floor.
12. The combination of claim 10, wherein the wheelchair docking system further comprises a bottom panel with a plurality of openings defined therein, the latch is coupled to the wheelchair docking system via one of the plurality of openings in the flange, and wherein the plurality of openings in the flange and the plurality of openings in the bottom panel are equally spaced and defined laterally to provide for a laterally adjustable coupling between the docking system and the track system.
13. The combination of claim 12, wherein each of the plurality of openings defined in the flange are longitudinally aligned with at least one of the plurality of openings defined in the bottom panel.
14. The combination of claim 10, wherein the latch comprises: a body with an opening defined therein; a tab portion being slidable in a generally vertical direction to releasably couple the latch to the track system; a first post and a second post; and a fastener coupled to the body via the defined opening, wherein the fastener is for coupling the latch to the wheelchair docking system.
15. The combination of claim 14, wherein the tab portion is adjustable between a raised position

and a lowered position, where in the lowered position the tab portion is disposed within a first receptacle of the plurality of receptacles of the at least one track, and in the raised position the tab portion is released from the first receptacle such that the wheelchair docking system and latch are longitudinally moveable relative to the track system.

16. The combination of claim 10, further comprising at least one retaining pin comprising a neck portion and a retaining end, the at least one retaining pin coupling the frame to the track system.

17. The combination of claim 16, wherein: the neck portion is aligned with and disposed within one of a plurality of openings in the frame and the channel of the at least one track; the retaining end is disposed within one of the receptacles of the plurality of receptacles.

18. The docking system of claim 1 in combination with the track system, wherein: the track system comprises a first track and a second track, the first and second tracks each including a first end, a second end, a plurality of receptacles, and an elongated channel defined therein between the first end and the second end; the wheelchair docking system is adjustably coupled to the track system; a first fastener adjustably coupling the wheelchair docking system to the first track, the first fastener being coupled to the frame; and a second fastener adjustably coupling the wheelchair docking system to the second track, the second fastener being coupled to the frame.

19. A wheelchair docking system for being coupled to a track system coupled to a vehicle floor, comprising: a frame; a coupler mechanism configured to engage a wheelchair as the wheelchair approaches the wheelchair docking system during a docking operation; an adjustable latch coupled to the frame, the adjustable latch configured to be releasably coupled to the track system; wherein the adjustable latch comprises a tab portion, a first post and a second post, the tab portion being slidable in a generally vertical direction to releasably couple the adjustable latch to the track system.

20. A wheelchair docking system for being coupled to a track system coupled to a vehicle floor, comprising: a frame; a coupler mechanism configured to engage a wheelchair as the wheelchair approaches the wheelchair docking system during a docking operation; an adjustable latch coupled to the frame, the adjustable latch configured to be releasably coupled to the track system; at least one retaining pin comprising a neck portion and a retaining end, the at least one retaining pin coupling the frame to the track system; wherein the neck portion is disposed within one of a plurality of openings in a bottom panel.
